Declarations

TechPulse Capstone Project - Case Study 1: AUTOSAR HLD Document Analysis Assistant (AutoDoc AI)

Student	Rithanyaa K
Register No.	127003212
Department	B.Tech Computer Science & Engineering
Institution	SASTRA Deemed University
Project	AutoDoc AI - AUTOSAR HLD Document Analysis Assistant (v2, local RAG)

Declaration 1 - Academic Integrity

I, Rithanyaa K (Register No. 127003212), declare that:

1. This submission is my original work, except where AI assistance is declared in Declaration 2.

2. All external code, libraries, datasets, models and documents are cited, including PyMuPDF, scikit-learn, pandas, Streamlit, sentence-transformers (all-MiniLM-L6-v2), FAISS (optional), Ollama, the hosted model openai/gpt-oss-20b accessed through an OpenAI-compatible API, and the TechPulse case-study document.

3. No results are fabricated. Reported results come from running the included scripts or from demonstrations clearly labelled as such in the technical report; items not tested or not measured are stated as such.

4. My individual contribution is accurately stated in the technical report and in Declaration 2.

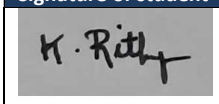
5. The demonstration documents are fictional synthetic test data and are not official AUTOSAR or Tata Technologies material.

Signature of student	Name	Date
	Rithanyaa K Register No. 127003212	20.10.2026

Declaration 2 - AI-Tool Usage

Every AI tool or pre-trained model used (including for coding assistance) is listed below. I remain accountable for correctness and can explain all submitted work.

AI Model/Tool	Purpose	Artifact affected	Verified by student
Claude (Anthropic)	Coding assistance: first drafts of the Python code (autodoc modules, Streamlit app, tests), synthetic-data generator, evaluation scripts, first documentation drafts	Code/*, Input_Data (synthetic generator), Documentation/*	Ran the unit tests (27 pass) and the app on my computer; read and edited the code; compared answers with the cited evidence
Claude (Anthropic) - later sessions	Source-code audit, correction and regeneration of the technical report, diagrams, documentation, declarations and packaging; small code changes (User-Agent header in llm.py, UI notice text in app.py, config defaults)	Documentation/*, Declarations/*, Code/autodoc/llm.py, Code/app.py, Model_Prompts_Config/config.json	Reviewed every report statement against the source code; re-ran the unit tests after the code changes; checked the report against my own demonstration
sentence-transformers/all-MiniLM-L6-v2	Run-time embedding model for passage and question embeddings	Vector index and retrieved evidence	Checked retrieved passages and similarity scores shown in the application
Ollama with phi:latest	Run-time local LLM used in the first demonstration to generate grounded answers	Answers shown in the application	Compared each answer with the displayed evidence passages [S1]-[S5]
openai/gpt-oss-20b (hosted via an OpenAI-compatible API, Groq)	Run-time hosted LLM used in the latest demonstration to generate grounded answers; only synthetic documents were sent	Answers shown in the application	Compared each answer with the displayed evidence passages and page citations

Signature of student	Name	Date
	Rithanyaa K Register No. 127003212	20.10.2026